

Midterm exam

1. Evaluate the definite integral

$$\int_0^3 u^2 e^{3u} du.$$

2. Evaluate the indefinite integral

$$\int \cos^4(3t) \sin^3(3t) dt.$$

3. Evaluate the improper integral or show that it diverges.

$$\int_{-\infty}^2 \frac{x}{x^2 + 4} dx.$$

4. Determine the length of the curve

$$y = \frac{5}{4}x + 1, \quad 0 \leq x \leq 8.$$

5. Find the surface area of the solid obtained by rotating the curve

$$y = 5 - 2x, \quad 0 \leq x \leq 3$$

about the y -axis.

6. Find the equation of the tangent line(s) to the parametric curve

$$x = 4 - e^{t^2 - 4t}, \quad y = t^3 - 6t^2 + 9t + 1$$

at the point $(3, 1)$.

Good Luck!